

even ground or even form themselves into solar sails attached to rockets, to unfurl in outer space and reduce the consumption of rocket fuel. They can help miners to explore the availability of rare minerals deep within the bellies of the planet. When they find something of interest, they can form themselves into an antenna and communicate the information to the base station.

More excitingly, NASA believes that the ANTS vehicular architecture would allow far more mobility for the physically-challenged than traditional wheelchairs. A high level of reconfiguration is possible, and there are no limitations such as those imposed by the fixed size of wheels and components in traditional vehicles. Relatively continuous motion is also possible.

With advanced command and control features, it would be possible to select a destination and sit back. The ANTS' addressable, self-configuring network of nodes would reversibly deploy struts, allowing transformation in size, shape and gait, as required and determined from feedback to the nodes at the actuator level.

NASA is not the only organisation working on swarms. A team at the University of Pennsylvania has developed fully-autonomous quadcopter drones that can build simple structures out of magnetic parts. The robots have the ability to work in swarms without interfering with each other. They also have the intelligence to know when a part is placed correctly and locked into place.

According to the researchers, their algorithm is limited only by the battery life of the quadcopters and the number of parts available. If deployed on a mass scale, this could make con-

struction of buildings a completely automated process.

Emerging field of technology

Nanorobotics is indeed an emerging field of technology, with constant research and development. Recent activity includes actual devices used in nanorobots, as well as systems to aid design and development, artificial intelligence algorithms and theories, and more. Here are some quick examples.

Chinese researchers have designed and tested simulations of a nano-clutch, which can be used to regulate the speed of nanomotors. The device consists of two carbon nanotubes, one inside the other, separated by a film of water.

The friction between the water and the walls of the carbon nanotubes is controlled using electro-wetting techniques. When the two tubes are electrically charged, the water between them can transmit the torque from the inner tube to the outer tube, and the device is said to be in the engaged state. When the carbon nanotubes are uncharged, the device is in the disengaged state.

According to a paper published by the inventors, the proposed device can perform step-less speed regulation by changing the magnitude of the charge assigned to the carbon nanotube atoms. The model may be helpful in designing and manufacturing nanorobots.

Researchers at the Monash Medical Centre, Melbourne, Australia, have recently developed a hardware architecture for developing nanorobotic systems for cancer therapy. The architecture includes support for nanobio-electronics, clinical data handling, embedded integrated circuits and devices, wireless technologies and so on.

Such and other hardware and software systems would speed up the process of nanorobotic systems development for various applications.

Big companies such as General Electric and Hewlett-Packard are researching and developing nanorobots, not to forget the heavily-funded research happening in several universities and medical centres across the world. Patents and patent litigation have also begun surfacing—which is more of a grey sign than a good one. Nanorobotics is an emerging field which requires heavy investment and contributions from multiple disciplines. So an open development model would be more fruitful than a closed, royalty-infested one—which often leads to one company monopolising the field and preventing large-scale benefits to the society.

The article on nanorobotics in the Wikipedia mentions that a document with a proposal on nanobiotech development using open technology approaches has been addressed to the United Nations (UN) General Assembly. According to the document sent to the UN, an approach similar to how Open Source accelerated the development of computer systems in recent years should be used to accelerate nanorobotics development also, to benefit society at large.

"The use of nano-biotechnology should be established as a human heritage for the coming generations, and developed as an open technology based on ethical practices for peaceful purposes. Open technology is stated as a fundamental key for such an aim," it reads.

"It is a fast evolving area, and the most detailed theoretical discussions on nanorobotics, including specific design issues such as sensing, power

KElement®




LED TV LCD 3D TV

POCKET SMART TV
Make Your Old TV to Internet TV



Tablet Pc's Models

Taupe
Woodbine
Purple Orchid
Endive
Bisque
Lilac

Price Starts From 3999/-

LEARNING SOLUTION for 21ST CENTURY



k tab®
www.ktab.in
admin@ktab.in
Trade Enquiry
099943 56789
080560 45640
1800 425 3099